

EXECUTIVE SUMMARY

This project, “**Nassau Candy Factory Reallocation & Shipping Optimization**,” focuses on improving factory allocation and shipping efficiency for a candy distribution operation using data analytics, predictive modeling, and scenario-based optimization. The primary objective is to identify suitable factory alternatives for different product, region, and shipping-mode combinations while reducing estimated delivery lead time and maintaining consideration of profitability and recommendation confidence.

The analysis was performed using a historical dataset containing **10,194 records** and **18 original operational variables**, including order information, product details, geographic information, shipping mode, sales, units, gross profit, and cost. During the data preparation stage, the dataset was cleaned and transformed to create analytical variables such as lead time, profit margin, sales per unit, profit per unit, order year, order month, order quarter, and order day of week. These variables were subsequently used to understand historical shipping patterns and support the factory optimization analysis.

The optimization framework evaluates factory alternatives across **15 products, 4 regions, and 4 shipping modes**, considering a total of **5 potential factories**. Scenario analysis was used to estimate the impact of reallocating products to alternative factories. The resulting analysis generated **770 scenario records** and **154 factory recommendations**. Each recommendation incorporates predicted lead time, lead-time reduction, speed-related scoring, profit-safety scoring, optimization score, and confidence level.

The results indicate that the proposed factory reallocation scenarios can achieve an **average estimated lead-time reduction of 20.0%** while maintaining an **average profit margin of approximately 54.77%** across the recommendation dataset. The recommendation framework produced **136 high-confidence recommendations and 18 medium-confidence recommendations**, with no low-confidence recommendations in the final recommendation dataset. The highest optimization score observed was approximately **0.971**, demonstrating that certain product-region-shipping combinations have particularly strong potential for factory reassignment.

A key component of the project is the interactive **Streamlit decision-intelligence dashboard**, which enables users to select a product, region, and shipping mode and immediately examine the corresponding factory scenarios. The dashboard presents current and recommended lead times, estimated lead-time reduction, confidence level, factory comparisons, what-if analysis, optimization scores, confidence distributions, risk information, and overall business impact. This allows the analytical results to be interpreted from an operational decision-making perspective rather than only through static statistical outputs.

The project also recognizes an important limitation of the analysis. The historical dataset does not contain observed shipments of the same product from multiple factories for every possible comparison. Therefore, alternative factory results represent **analytical simulations based on historical product-region-shipping performance and assumed factory scenario factors**, rather than direct observations of actual factory transfers. Consequently, the recommendations should be considered decision-support outputs and should be validated

against actual factory capacity, inventory availability, transportation costs, service-level requirements, and operational constraints before implementation.

Overall, the project demonstrates how **data cleaning, exploratory analysis, predictive modeling, scenario analysis, optimization scoring, and interactive visualization** can be integrated into a practical decision-support system. The resulting solution provides a structured approach for identifying potential factory reallocation opportunities, evaluating their expected impact on shipping lead time, and supporting more informed operational planning for Nassau Candy Distributor.

Project Metric	Result
Historical Records	10,194
Products	15
Regions	4
Ship Modes	4
Potential Factories	5
Scenario Records	770
Factory Recommendations	154
Average Lead-Time Reduction	20.0%
Average Profit Margin	54.77%
High-Confidence Recommendations	136
Medium-Confidence Recommendations	18